Research Notes on Ridge regression

Ridge regression

$$L(\beta) = \|y - X\beta\|^2 + \lambda\|\beta\|^2$$

$$\hat{\beta} = (X^T X + \lambda I)^{-1} X^T y$$

>> Section 1

and is zero elsewhere. This demonstrates the effect of the Tikhonov parameter on the condition number of the regularized problem. For the generalized case, a similar representation can be derived using a generalized singular-value decomposition. Finally, it is related to the Wiener filter:

>> Section 2

Plug-in estimator Assume that X {\displaystyle \mathbf {X} } is an n × p {\displaystyle n\times p} matrix and define the matrix Ω := (X ■ X / n) + {\displaystyle \Omega :=(\mathbf {X} ^{\top }\mathbf {X} /n)^{+}} . Then, consider the following choice for the Tikhonov regularization parameter:

>> Section 3

Further reading Gruber, Marvin (1998). Improving Efficiency by Shrinkage: The James–Stein and Ridge Regression Estimators. Boca Raton: CRC Press. ISBN 0-8247-0156-9. Kress, Rainer (1998). "Tikhonov Regularization". Numerical Analysis. New York: Springer. pp. 86–90. ISBN 0-387-98408-9. Press, W. H.; Teukolsky, S. A.; Vetterling, W. T.; Flannery, B. P. (2007). "Section 19.5. Linear Regularization Methods". Numerical Recipes: The Art of Scientific Computing (3rd ed.). New York: Cambridge University Press. ISBN 978-0-521-88068-8. Saleh, A. K. Md. Ehsanes; Arashi, Mohammad; Kibria, B. M. Golam (2019). Theory of Ridge Regression Estimation with Applications. New York: John Wiley & Sons. ISBN 978-1-118-64461-4. Taddy, Matt (2019). "Regularization". Business Data Science: Combining Machine Learning and Economics to Optimize, Automate, and Accelerate Business Decisions. New York: McGraw-Hill. pp. 69–104. ISBN 978-1-260-45277-8.

>> Section 4

In the context of arbitrary likelihood fits, this is valid, as long as the quadratic approximation of the likelihood function is valid. This means that, as long as the perturbation from the unregularised result is small, one can regularise any result that is presented as a best fit point with a covariance matrix. No detailed knowledge of the underlying likelihood function is needed.

>> Section 5

the problem of a near-singular moment matrix X T X {\displaystyle \mathbf {X} ^{\mathsf {T}}\mathbf {X} } is alleviated by adding positive elements to the diagonals, thereby decreasing its condition number. Compared to the ordinary least squares estimator, the simple ridge estimator has an extra term λ I {\displaystyle \lambda \mathbf {I} } in the denominator:

>> Section 6

where Y {\displaystyle \mathbf {Y} } is the regressand or response vector, X {\displaystyle \mathbf {X} } is the design matrix, I {\displaystyle \mathbf {I} } is the identity matrix, and the ridge (or Tikhonov) regularization parameter λ ≥ 0 {\displaystyle \lambda \geq 0} serves as the constant shifting the diagonals of the moment matrix. It can be shown that this estimator is the solution to the least squares problem subject to the constraint β T β = c {\displaystyle {\boldsymbol {\beta }}^{\mathsf {T}}{\boldsymbol {\beta }}=c} , which can be expressed as a Lagrangian minimization:

>> Section 7

The Lavrentyev regularization, if applicable, is advantageous to the original Tikhonov regularization, since the Lavrentyev matrix A + Q {\displaystyle A+Q} can be better conditioned, i.e., have a smaller condition number, compared to the Tikhonov matrix A T A + Γ T Γ . {\displaystyle A^{\mathsf {T}}A+\Gamma ^{\mathsf {T}}\Gamma .}

>> Section 8

Generalized cross-validation estimator A common data-driven choice for λ {\displaystyle \lambda } is the minimizer of the cross-validation loss or its generalizations. For example, Grace Wahba proved that the optimal parameter, in the sense of generalized cross-validation minimizes

>> Section 9

with the "regularisation matrix" B = (A T A + Γ T Γ) − 1 A T A {\displaystyle B=\left(A^{\mathsf {T}}A+\Gamma ^{\mathsf {T}}\Gamma \right)^{-1}A^{\mathsf {T}}A} . If the parameter fit comes with a covariance matrix of the estimated parameter uncertainties V 0 {\displaystyle V_{0}} , then the regularisation matrix will be

>> Section 10

Regularization in Hilbert space Typically discrete linear ill-conditioned problems result from discretization of integral equations, and one can formulate a Tikhonov regularization in the original infinite-dimensional context. In the above we can interpret A {\displaystyle A} as a compact operator on Hilbert spaces, and x {\displaystyle x} and b {\displaystyle b} as elements in the domain and range of A {\displaystyle A} . The operator A ∗ A + Γ T Γ {\displaystyle A^{*}A+\Gamma ^{\mathsf {T}}\Gamma } is then a self-adjoint bounded invertible operator.

>> Section 11

Tikhonov regularization for linear equations Suppose that for a known real matrix A {\displaystyle A} and vector b {\displaystyle \mathbf {b} } , we wish to find a vector x {\displaystyle \mathbf {x} } such that

>> Section 12

where the expectations treat X {\displaystyle \mathbf {X} } as fixed and Y ′ {\displaystyle \mathbf {Y} '} is test response data, independent from Y {\displaystyle \mathbf {Y} } (and hence independent from the estimators β ^ 0 {\displaystyle {\hat {\boldsymbol {\beta }}}_{0}} and β ^ λ ∗ {\displaystyle {\hat {\boldsymbol {\beta }}}_{\lambda ^{*}}}). Of course, in practice the formula for λ ∗ {\displaystyle \lambda ^{*}} is used by plugging in statistical estimators for the unknown parameters β {\displaystyle {\boldsymbol {\beta }}} and ς 2 {\displaystyle \varsigma ^{2}} . When n > p {\displaystyle n>p} , the most natural estimators for these parameters are the usual least-squares ones:

>> Section 13

β ^ λ = (X T X + λ I) − 1 X T Y {\displaystyle {\hat {\boldsymbol {\beta }}}_{\lambda }=\left(\mathbf {X} ^{\mathsf {T}}\mathbf {X} +\lambda \mathbf {I} \right)^{-1}\mathbf {X} ^{\mathsf {T}}\mathbf {Y} }

>> Section 14

Generalized Tikhonov regularization For general multivariate normal distributions for x {\displaystyle \mathbf {x} } and the data error, one can apply a transformation of the variables to reduce to the case above. Equivalently, one can seek an x {\displaystyle \mathbf {x} } to minimize

>> Section 15

where A = (A Γ) {\displaystyle {\mathcal {A}}={\begin{pmatrix}A\\\Gamma \end{pmatrix}}} and b = (b 0) {\displaystyle {\mathcal {b}}={\begin{pmatrix}\mathbf {b} \\{\boldsymbol {0}}\end{pmatrix}}} , for some suitably chosen Tikhonov matrix Γ {\displaystyle \Gamma } . In many cases, this matrix is chosen as a scalar multiple of the identity matrix (Γ = α I {\displaystyle \Gamma =\alpha I}), giving preference to solutions with smaller norms; this is known as L2 regularization. In other cases, high-pass operators (e.g., a difference operator or a weighted Fourier operator) may be used to enforce smoothness if the underlying vector is believed to be mostly continuous. This regularization improves the conditioning of the problem, thus enabling a direct numerical solution. Treating it as an ordinary least squares problem with augmented matrices A {\displaystyle {\mathcal {A}}} and b {\displaystyle {\mathcal {b}}} , the solution is